Operations Research 1: Deterministic Models
Course Introduction

Dr. Lei Zhao

Department of Industrial Engineering

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Operations & Services Research (TOpS) Laboratory
Department of Industrial Engineering
Tsinghua University, China

About Us

- Instructor
  - Dr. Lei ZHAO (赵磊)
    - Office: Shunde Bldg. (舜德楼), Rm. 408
    - Mobile: 139 1121 2264
    - Email: lzhao@tsinghua.edu.cn
    - Office hours: by appointment

About Us

- Teaching assistants
  - Tirui CAO (曹体睿)
    - Office: Shunde Bldg. (舜德楼), Rm. 325E
    - Mobile: 185 0268 8881
    - Email: ctr20@mails.tsinghua.edu.cn
    - Office hours: 14:00 – 15:00, Mondays (by appointment)

  - Yiru WANG (王艺如)
    - Office: Shunde Bldg. (舜德楼), Rm. 324
    - Mobile: 177 7960 8339
    - Email: yr-wang23@mails.tsinghua.edu.cn
    - Office hours: 19:00 – 20:00, Fridays (by appointment)

About Us

- Teaching assistants
  - Xudong LUO (罗旭东)
    - Office: Shunde Bldg. (舜德楼), Rm. 325E
    - Mobile: 188 1061 9869
    - Email: luoxd24@mails.tsinghua.edu.cn
    - Office hours: 14:00 – 15:00, Thursdays (by appointment)

  - Keyi YAN (严珂一)
    - Office: Shunde Bldg. (舜德楼), Rm. 325E
    - Mobile: 135 5853 3001
    - Email: yan-ky24@mails.tsinghua.edu.cn
    - Office hours: 15:30 – 16:30, Mondays (by appointment)

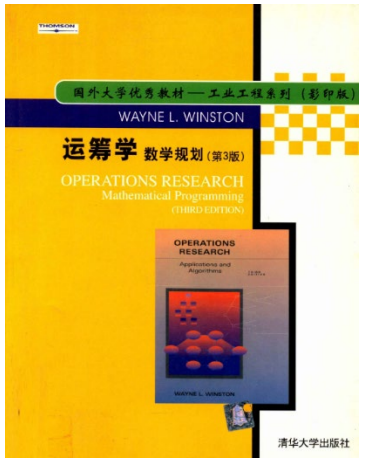
About Us

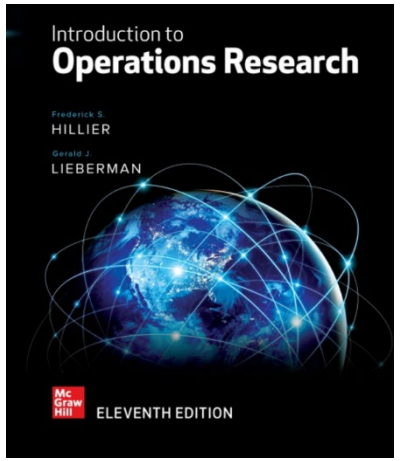
- Teaching assistants
  - Zhonghao CHANG (常中豪)
    - Office: Shunde Bldg. (舜德楼), Rm. 325E
    - Mobile: 139 3939 3873
    - Email: changzh25@mails.tsinghua.edu.cn
    - Office hours: 17:00 – 18:00, Wednesdays (by appointment)

  - Tingan LIN (林廷安)
    - Office: Shunde Bldg. (舜德楼), Rm. TBD
    - Mobile: 139 5869 1763
    - Email: lta25@mails.tsinghua.edu.cn
    - Office hours: 10:30 – 11:30, Wednesdays (by appointment)

About the Course

- Operations Research 1: Deterministic Models
  - Course ID: 30160023
  - Time: 13:30 – 16:05, Wednesdays
  - Location: Shunde Bldg. (舜德楼), Rm. 401
- Prerequisites
  - Calculus, linear algebra, computer programming
- References
  - Winston, Wayne L. (2004). *Operations Research: Mathematical Programming (运筹学：数学规划)*, 3rd edition, Thomson Learning (authorized reprint edition by Tsinghua University Press).
  - Hillier, Frederick S. and Gerald J. Lieberman (2021). *Introduction to Operations Research*, 11th edition, McGraw Hill.





About the Course

- Course Description
  - Deterministic optimization models and algorithms, including
    - Linear programming, integer programming, nonlinear programming, and
    - Dynamic programming

- Course Objectives
  - Learn modeling techniques to translate (practical) optimization problems into mathematical optimization models
  - Learn fundamental optimization algorithms to solve the optimization models efficiently
  - Learn to use computer software to solve (relatively) large-scale optimization problem instances

About the Course

- Tentative course topics
  1. Introduction to operations research
  2. Linear programming (LP)
     a) Basic concepts
     b) Modeling examples
     c) Simplex methods
     d) Sensitivity analysis
     e) Duality
  3. Integer programming (IP)
     a) Modeling examples
     b) Modeling techniques
     c) Branch and bound methods
     d) Cutting plane methods
     e) Advanced modeling examples
  4. Nonlinear programming (NLP)
     a) Convex sets and convex functions
     b) Unconstrained optimization
     c) Constrained optimization (tentative)
  5. Dynamic programming (DP)
     a) Deterministic finite state finite horizon problem
     b) Principle of optimality
     c) Dynamic programming algorithm
     d) Stochastic finite state finite horizon problem (optional)

About the Course

- Announcements (网络学堂)
- Lecture files / References (网络学堂)
- Homework (网络学堂)
  - Due date: One week after the assignment date
  - Due time: Beginning of the class
  - Late penalty
    - < 1 day: 10% off
    - < 2 days: 20% off
    - >= 2 days: 100% off
  - Hall of fame: +5%
  - **Plagiarism: Strictly prohibited (0%)!!!**

About the Course

- Class participation: Required
  - Classroom (教室) & Online (雨课堂)
- Course project: No
- Grading policy

Homework	20%
Midterm exam	30%
Final exam	40%
In-class quizzes	5%
In-class participation	5%
Total	100%

Q & A